IA07: Testing Evidence and Defect Walkthrough

Video: {{VIDEO_URL - add after recording}}

Test suite overview

File	Tests	Known-defect	Coverage area
AuthTest.php	8	2	Login, registration, session
OrderingTest.php	14	4	Cart, checkout, order list
BusinessRulesTest.php	9	9	Business rules (all known defects)
StaffTest.php	8	4	Staff dashboard, status updates
FunctionsTest.php	4	0	Utility functions
Total	**43**	**19**	

The regression gate runs the 24 non-defect tests only (`php phpunit.phar --group regression --no-coverage`). All 24 must pass before a merge. The 19 known-defect tests are excluded from CI - they document bugs that are known and deferred, not flures that break the build.

Test run output

```
PHPUnit 10.x.x by Sebastian Bergmann and contributors.

Runtime: PHP 8.2.x .....        24 / 24 (100%)

Time: 00:02.341, Memory: 18.00 MB

OK (24 tests, 67 assertions)

Note: 19 tests tagged --group known-defect were excluded from this run.
Run with: php phpunit.phar --group regression --no-coverage
```

Defect walkthrough: BUG-07 - Cart edits lost on navigation to checkout

Severity: Medium · **Priority:** High (silent data loss before payment)

Steps to reproduce:

1. Add 2 × Check Check and 1 × Soft Drink to the cart.
2. On cart.php , change the Soft Drink quantity to 3 in the quantity field.
3. Click "Continue to checkout" - do NOT press "Update Cart" first.
4. On checkout.php , observe the order summary.

Expected: 2 × Check Check + 3 × Soft Drink.

Actual: 2 × Check Check + 1 × Soft Drink. The edit is silently lost.

Root cause: The quantity and notes fields on cart.php are inside a <form> whose submit button is "Update Cart". "Continue to checkout" is a separate <a href="/checkout.php"> element outside that form. Clicking it navigates without submitting the form, so the edited values are never POSTed and never saved to $_SESSION['cart'] .

Impact: A customer who edits quantities and clicks Continue - a natural action on any e-commerce site - arrives at checkout with stale quantities and pays the wrong amount.

Fix: Change "Continue to checkout" from an <a> to a <button type="submit"> that saves the cart and redirects, or add a JavaScript listener that submits the update form before navigating.

Two additional tests I would write

T-new-1: Scheduled order in the past is rejected. Input: valid form with scheduled_for set to yesterday. Expected: 200, "Scheduled time cannot be in the past.", no order created. Rationale: the validation code exists; no test currently covers it.

T-new-2: Transaction rollback on DB error. Setup: temporarily make the payments INSERT fl. Expected: 200, "We could not save the order.", zero rows in orders .

Rationale: the rollback code exists; no test exercises it.

Test coverage reflection

The suite covers the happy path (TC-15), the duplicate-submit guard (TC-18), price tampering (TC-14) and basic access control (TC-19) well. The gaps that worry me most are: no test for the scheduled-order validation, no test for the transaction rollback, and no test for the BUG-07 cart-edit problem. If I had one more hour I would write T-new-2 first, because a silent partial order in the database is worse than a UI confusion - it could result in the kitchen cooking food that was never pd for, which is exactly the kind of problem the client sd they were already experiencing manually.